Design and Analysis of Axial Turbine Power Extraction from a Small-Scale Rotating Detonation Rocket Combustor

The Rotating Detonation Turbine Generator (RDTG) project, in conjunction with Braun's Engineering Factory for Advanced Supersonic Technologies (BEFAST Lab) at North Carolina State University, aims to present an experimental set-up to prove the feasibility of high-efficiency power generation using an axial turbine generator in the exhaust of a rotating detonation engine (RDE). This is accomplished through the design, optimization, and experimental validation of a small turbine blisk capable of extracting power from BEFAST Lab's RDE exhaust while handling the adverse effects of detonation-induced turbulence and thermal gradients. RDE technology offers thermodynamic advantages over conventional deflagration-based systems. However, the inherently unsteady, supersonic exhaust flow presents challenges for a typical turbine integration, including extreme pressure fluctuations, shock-induced energy losses, and thermal stresses that can compromise structural integrity and operational articles placed downstream of the exhaust. The small-scale turbine rotor-stator assembly is designed with consideration of detonation induced shockwave, incorporating exit area reductions to increase upstream pressure and reduce axial Mach number experienced by the turbine stage. Blade geometry design is conducted using Advanced Design Technology and Aerodynamic Solutions software, ensuring optimal aerodynamic efficiency. CFD simulations using ANSYS Fluent, ANSYS CFX, and ADS CFD are employed to refine turbine design, with an emphasis on aerodynamic efficiency and structural soundness. Structural and dynamic predictions for the rotor, shaft, and bearings are conducted using MATLAB as well as Simscape tools, which allows for the determination of shaft resonance, shaft deflection, and bearing loading to drive shaft, bearing, and coupling design and selection. Furthermore, finite element methods are used in ANSYS Mechanical to assess material stresses and ensure durability under extreme thermal loads for exhaust routing systems, the motor mount interface, and shaft-rotor system. The proposed experimental validation involves controlled short-duration RDE firings; a target electrical power output of 3 kW serves as the primary performance benchmark, while secondary objectives encompass efficiency maximization, thermal stability, and mechanical longevity. Instrumentation, including a high-frequency pressure transducer and pressure taps, thermocouples, and torque sensors, allows real-time data acquisition of flow properties, thermal conditions, and mechanical performance during testing. This study represents a proof in concept in realizing compact, power-dense, and thermodynamically optimized energy systems capable of operating under extreme flow and combustion conditions, and will act as the first step in BEFAST Lab's platform for developing RDE-driven axial turbines.